A dynamic model of nutrient pathways, growth, and body composition in fish

Nadav S. Bar, Trygve Sigholt, Karl D. Shearer, and Åshild Krogdahl

Abstract: The growth and body composition of an organism are dynamic and depend on available diet, as well as other environmental variables. A structured model is described predicting growth, body composition, and the first limiting amino acid for a variety of feeding regimes over a relatively long time scale. This model continuously provides, as a function of time, the flow of nutrients and metabolites through the principal metabolic pathways leading to tissue growth. Measurements of growth and whole body composition (including amino acid composition) from a large-scale experiment with Atlantic salmon (*Salmo salar*) have been employed to calibrate this model. Comparisons of simulations with the results of feeding experiments validate the predictive ability of the model. These types of simulations will be valuable in studying the quantitative relationships between intracellular energy levels and nutrient distribution in tissues. They will also provide data for balancing diet composition and amino acid profile in order to optimize growth with respect to factors such as body weight, body composition, and ambient temperature. As a result of its modularity, the model can be easily extended to include additional physiological and metabolic processes.

Résumé : La croissance et la composition corporelle d'un organisme sont des processus dynamiques qui dépendent du régime alimentaire disponible, ainsi que d'autres variables du milieu. Nous décrivons un modèle structuré qui prédit la croissance, la composition corporelle et le premier acide aminé limitant pour une variété de régimes alimentaires sur une période de temps relativement longue. Le modèle fournit de manière continue en fonction du temps le flux des nutriments et des métabolites en suivant la voie métabolique principale qui mène à la croissance des tissus. Pour calibrer notre modèle, nous utilisons des mesures de croissance et de composition corporelle globale (incluant la composition en acides aminés) provenant d'une expérience à grande échelle menée sur le saumon atlantique (*Salmo salar*). Une comparaison des simulations et des résultats d'expériences alimentaires permet de valider la capacité de prédiction du modèle. Les simulations de ce type sont précieuses pour étudier les relations quantitatives entre les niveaux d'énergie intracellulaire et la répartition des nutriments dans les tissus. Elles fournissent aussi des données pour équilibrer la composition des régimes alimentaires et les profils d'acides aminés de façon à maximiser la croissance en relation avec des facteurs tels que la masse corporelle, la composition corporelle et la température ambiante. Parce qu'il est modulaire, notre modèle peut facilement être élargi pour inclure des processus physiologiques et métaboliques additionnels.

[Traduit par la Rédaction]

Introduction

In the last few decades, much effort has been invested in developing a mathematical description of fish nutrition and growth in an attempt to optimize feeds. For instance, Hua and Bureau (2006) estimated the digestible phosphorus content of salmonid feeds using multiple regression. They reported that their model is valid over a wide variety of ingredients commonly used in fish feeds, despite its simplicity. Cacho (1990) and Hop et al. (1997) used curve-fitting methods and data extracted from various experiments to construct bioenergetic models for channel catfish (*Ictalurus punctatus*) and Arctic cod (*Boreogadus saida*). This class of models expresses growth in a relatively simple manner, allowing experimental comparison of different strains and (or) feeding regimes. Their models do not require comprehensive understanding of the processes involved. In addition, these models are based largely on data sets derived from multiple experiments conducted under specific conditions. These models can only be applied with confidence within a relatively small range of conditions and, more specifically, with data of a background similar to those used to derive their equations. To address this issue for pigs, cattle, and chickens, several dynamic models have been developed (Vetharaniam et al. 2001*a*, 2001*b*; MacNamara 2003). These models effectively predict growth and body composition. Few models based on a similar approach have been used for fish. Jørgensen and Fiksen (2006) simulated growth of larval

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N.S. Bar.[1] Department of Engineering Cybernetics, Norwegian University of Science and Technology (NTNU), NO-7491 Trondheim, Norway.
T. Sigholt. BioMar AS, NO-7484 Trondheim, Norway.
K.D. Shearer. National Marine Fisheries Service, 2725 Montlake Boulevard E., Seattle, WA 98112, USA.
Å. Krogdahl. Aquaculture Protein Centre, Norwegian School of Veterinary Science, NO-0033 Oslo, Norway.

[1]Corresponding author (e-mail: barc@itk.ntnu.no).

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Atlantic cod (*Gadus morhua*) in a mechanistic manner. Conceiçâo et al. (1998) developed a simplified dynamic model for simulation of growth in fish larvae. They reported successful validation of the model in African catfish (*Clarias gariepinus*) and turbot (*Scophthalmus maximus*) for a short time span of 10–20 days. A model of fish growth in response to ecophysiological factors, based on bioenergetics, has shown consistency in a variety of enviromental conditions (Neill et al. 2004). Olsen and Balchen (1992) attempted to model fish physiology in a structured manner using differential equations to describe the main metabolic pathways of fish in general. However, because of an insufficient level of modeling details, it was impossible to predict important characteristics such as the first limiting amino acid, and therefore, their model is not adequate for the purpose of feed design.

The emerging field of systems biology integrates molecular and cellular biology with systems theory (Wolkenhauer et al. 2005). A new trend in mathematical modeling has emerged by focusing on the so-called metabolic flux analysis (Provost and Bastin 2004) and dynamic energy and mass budgets (Koojiman 2000). This field complements ongoing research in fish nutrition because of the potential to apply new modeling tools and theories to the extensive data from nutritional studies that exist today.

We present a deterministic dynamic model that takes into account flow of nutrients and metabolites, rates of metabolic processes, and energy budget as functions of time. The model belongs to a class of nonlinear systems

$$(1a) \qquad \dot{x} = f(x, u; t)$$

$$(1b) \qquad y = h(x; t)$$

where $\dot{x} = \mathrm{d}x_i/\mathrm{d}t$ is the rate of the variable i, f denotes a nonlinear system of equations, and t is time. The term u represents the input variables such as feed ingredients; x relates to system variables such as amino acids, protein level, and tricarboxylic acid (TCA) cycle intermediates; y represents the variables that can be measured, denoted output variables, such as growth and body composition; and h is the function describing the manner in which these variables are measured. This framework is widely used in kinetic models of metabolic networks, and extensive discussions can be found in Segel (1984), Britton (2003), and Gerdtzen et al. (2004). The work of Olsen and Balchen (1992), mentioned earlier for fish, is also based on the above representation. The model was implemented using MATLAB and C programs. The complete set of equations is available in Bar (2007).

The primary objective of the model is to predict growth, body composition, and the first limiting amino acid under a variety of conditions and feeding regimes and over a relatively long time scale. The model can also contribute to improve our understanding of metabolism in Atlantic salmon. In other words, the use of a structured mathematical model combined with empirical data enables the scientist to investigate various feeding scenarios, thus providing tools for hypothesis testing. Whether one employs static models or complex dynamic models, it is not always possible to estimate all necessary parameters from experimental data. Different strategies are applied to overcome this difficulty (Wolkenhauer et al. 2005). We emphasize that in most cases, it is not a question of choosing either modeling or empirical work; rather, empirical data and modeling are parts of an integrated process. As the authors in Cho and Wolkenhauer (2003) discuss in the framework of system biology, even if we are unable to build accurate predictive models of fish growth, the modeling process itself may prove valuable to nutritionists, helping to identify which variables need to be measured and why.

This article is structured as follows. Methods presents the key elements of the metabolic networks used in the model followed by Results, simulating a case study reported in the literature. This validation of the model is the essence of the work. In this article, the notation k_i will denote rate constant of reaction i, and u_i will denote a regulated rate of reaction i. A complete list of variables and parameters used in the present model is provided by Tables 1 and 2.

Methods

The metabolic model

General description and assumptions

The metabolic model (Fig. 1) is based on mass action kinetics of some of the most substantial processes in the body. The model covers the portion of the life cycle with rapid growth — between the juvenile stage and sexual maturity. During this period, the growth dynamic is not influenced by large hormonal changes, e.g., those that occur during sexual maturation. Accumulated energy is utilized to fuel other processes and regulates flow of the metabolic pathways in a nonlinear manner. The primary components of the model include protein and lipid metabolism and the TCA cycle, which functions as the “core” of the metabolic model. Nutrients are fed into the model using a two-stage Michaelis–Menten model for digestion and absorption of nutrients. Stoichiometric reactions are translated into differential equations, expressed as indicated by eqs. 1*a* and 1*b*, that describe the change of variables with time (a list of the variables used in the model is in Table 1).

As mentioned earlier, some assumptions are necessary to obtain a working model. (*i*) The apparent digestible coefficient (ADC) of the nutrients is assumed known or can be acquired from experiments. (*ii*) The supply of nonessential amino acids is abundant and the model only incorporates synthesis of cysteine and alanine. (*iii*) The fish consists of a single homogeneous protein that defines a certain amino acid composition of the species. This assumption simplifies the model as it integrates different tissue compositions into a single generic protein. Tissues have different rates of protein synthesis, which can vary by 15-fold, e.g., the case of kidney and white muscle of Atlantic cod (Carter and Houlihan 2001). Because muscle is approximately 60% of the total mass in salmon, this assumption is sufficient for the purpose of the model. (*iv*) The amino acid requirements for protein tissue growth was given by Bendiksen et al. (2006). Amino acid requirements for maintenance proteins (e.g., gut and immune system related proteins with high turnover rates) were developed from Halver and Hardy (2002). (*v*) Tissues maintain levels of adenosine triphosphate (ATP) within limited bounds (Caldwell and Hinshaw 1994; Zietara et al. 2004) through complex regulatory mechanisms (Lehninger 2005).

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Table 1. Variables and rates used in the present model.

Notation	Units	Details
oxal	nmol	Oxaloacetate concentrations
Acc	nmol	Acetyl-CoA concentrations
citr	nmol	Citrate concentrations
α-keto	nmol	α-Ketoglutarate concentrations
succ	nmol	Succinate concentrations
fum	nmol	Fumarate concentrations
mal	nmol	Malate concentrations
pyr	nmol	Pyruvate concentrations
ATP, ADP	nmol	Levels of ATP and ADP in the cells
NADH, NAD+	nmol	Levels of NADH and NAD in the cells
palm	nmol	Palmitate concentrations
malonyl-CoA	nmol	Malonyl-CoA concentrations
TAG	nmol	Triacylglycerol concentrations
FA	nmol	Fatty acids concentrations
AA	nmol	(Vector) concentrations of 20 amino acids
Glc	nmol	Glucose concentrations
O_2	nmol	Oxygen concentrations
NH_3	nmol	Ammonia concentrations
p_b	μmol·g dry weight^{-1}	Protein mass in the body
p_d	μmol·g dry weight^{-1}	Degraded protein mass
p_s	μmol·g dry weight^{-1}	Synthesized protein mass
u_i	nmol/min	Control rate of the *i* reaction
v_d	nmol/min	Rate variable of protein degradation
m_w	g	Total wet mass of the body
dw	g	Total dry mass of the body
w	g	Total water content in the body
γ	mg/h	Slow-varying variable due to age and life-history stage

Amino acids pathways

Amino acids (AA) are utilized both as a source of energy in the body and as building blocks for peptide chains; thus the pathways involving amino acids are central for predictions in the present model. They are transported with the blood from the intestines to the liver where some are modified before further transport to the organs and tissues for further utilization. Amino acids cross the cell membranes by active transport and are used in the protein synthesis process or oxidized. In cases of amino acid deficiencies, rate of protein synthesis is suppressed and amino acids are consequently diverted towards catabolism. The latter process involves transamination and oxidative deamination, which releases NH_3, described by the following reactions:

$$\text{(2)}\qquad \mathrm{AA} + \alpha\text{-keto} \rightarrow \alpha\text{-ketoglutaric acid} + \text{glutamate}$$

$$\text{(3)}\qquad \text{glutamate} + \mathrm{NAD}^{+} + \mathrm{H_2O} \rightarrow \alpha\text{-keto} + \mathrm{NH_3} + \mathrm{NADH}$$

The liver is the main organ for these reactions in fish, as in other animals, but also in fish, the skeletal muscle appears to liberate quantitatively important amounts of ammonia to the blood stream. Reaction 3 (eq. 3) is oxidative deamination of glutamate dehydrogenase (GDH), which is promoted during periods when ATP levels are high (Ballantyne 2001). The enzymes histidase, asparginase, serine dehydratase, threonine dehydratase, and glutaminase, as well as GDH, deaminate their respective amino acids without transamination. The breakdown rate of amino acids to ketone bodies varies between tissues and amino acids; however, these variations are negligible in the growth model because of the short reaction time compared with the time needed for tissue accretion. Energy released during catabolism is channeled through the electron transport system linked to the recycling of NAD^+ and coupled to ATP synthesis (details in the following sections).

TCA pathways

The TCA cycle functions as an energy source and distributor of intermediates to other processes in this model. Several intermediates in the TCA cycle are not included as they have only one pathway and thus can be regarded as temporal transition phases, e.g., isocitrate. The remaining reactions are mostly irreversible (Fig. 2).

The assumption of similar rate constants k_{TCA} for several of the reactions in the TCA cycle is reasonable as no other control mechanism is introduced (Voet 2004); therefore, choosing the average kinetic value of these reactions will serve the purpose of the model. Oxaloacetate exists in the cell in low concentrations as $k_{mal} \gg k_{TCA}$ and is rapidly picked up by citrate synthase and condensed with acetyl-CoA (Acc) to citrate (Lehninger 2005). Availability of the latter enzyme is controlled by the ratio of ADP to ATP in the cells such that accumulation of ATP inhibits citrate synthase, suppressing the flow of acetyl-CoA to the cycle. Several amino acids enhance levels of TCA cycle intermediates upon breakdown, including oxaloacetate. This may risk impairing the consistent operation of the TCA cycle and lead to erroneous energy balance. However, a controlled rate of malate transformation to

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Table 2. Parameters and their values used in the present model.

Parameter	Description	Value	Source
α_{acc}	Steady state value of acetyl-CoA in u_{acc} control	0.9 $\mu mol \cdot m_w^{-1}$	Salmon[1]
β_{acc}	Deviation power from steady state value of u_{acc}	3–4	Assumed
α_{oxal}	Steady state value of oxaloacetate in u_{oxal}	0.04 $\mu mol \cdot m_w^{-1}$	Salmon[1]
β_{oxal}	Deviation power from steady state value of u_{oxal}	2	Assumed
k_{oxal}	Kinetic rate of oxaloacetate in TCA	$0.3 \cdot s^{-1}$	Characidae[2]
α_{ls}	Mean steady state value of ATP in u_{ls} control	10 $\mu mol \cdot m_w^{-1}$	Fish[3,4,5]
β_{ls}	Deviation power from steady state value of u_{ls}	4	Assumed
α_{ld}	Lower bound of ATP at steady state in u_{ld} control	7 $\mu mol \cdot m_w^{-1}$	Fish[3,4,5]
β_{ld}	Deviation power from steady state value of u_{ld}	8	Assumed
α_{xd}	Low value of ATP in u_{xd} control	8 $\mu mol \cdot m_w^{-1}$	Fish[4,6,7]
β_{xd}	Deviation power from steady state value of u_{xd}	8	Assumed
α_{citr}	Normal ATP operation value for enzyme citrate synthase	10 $\mu mol \cdot g^{-1}$	Fish[3,4,5,8]
k_{mal}	Reaction rate of malate	$0.23 \cdot min^{-1}$	Trout[4]
k_{ls}	Kinetic rate of palmitate formation	0.75 $nmol \cdot g^{-1} \cdot min^{-1}$	Trout[9]
k_{TAG}	Kinetic rate of TAG formation	8.1 $nmol \cdot g^{-1} \cdot min^{-1}$	Trout[8]
k_{FA}	Kinetic rate of FA formation	4 $nmol \cdot g^{-1} \cdot min^{-1}$	Trout[8]
k_{TCA}	Kinetic rate TCA cycle	0.5 $nmol \cdot g^{-1} \cdot min^{-1}$	Calculated[10]
$k_{activity}$	Rate of ATP consumption in muscle due to activity	1–3 $\mu mol \cdot m_w^{-1} \cdot h^{-1}$	Trout, salmon[3,4,17]
$k_{essential}$	Rate of ATP consumption due to essential processes	10 $\mu mol \cdot m_w^{-1} \cdot h^{-1}$	Trout[17,18]
T_{op}	Optimal operation temperature (°C)	13 °C	Salmon[11]
ψ	Genetically determined efficiency of rate of protein degradation	0.2 ~ 0.85	Assumed
k_e, k_{-e}	Kinetic rate of transformations NADH-NAD+	0.5, 0.01$\cdot min^{-1}$	Estimated[12]
f_p, f_t	Flux of amino acids postdegradation process	0.75, 0.25	Estimated[13,14]
ζ	Fraction of minerals in the dry body weight	0.015	Salmon[15]
ρ_L	Fraction of water displaced when lipid is added to the adipose	0.3–0.9	Salmon[15]
ρ_p	Fraction of water associated with protein	3.8–4	Salmon[15,16]

Note: Sources: 1, taken from data on Atlantic salmon from Moya-Falcón et al. (2006); 2, taken from experiments on *Astyanax fasciatus* reported by de Luca et al. (1983); 3, estimated from experiments on juvenile salmon in Waiwood et al. (1992); 4, estimated from Mommsen and Hochachka (1988) for trout white muscle; 5, estimated from Lushchak and Storey (1994) for trout white muscle; 6, taken from data experiments on rainbow trout reported by Caldwell and Hinshaw (1994); 7, taken from experiments on flounder (Jørgensen and Mustafa 1980); 8, estimated from experiments on goldfish reported by Thillart et al. (1976); 9, taken from data reported by Bernard et al. (1999) for rainbow trout; 10, Voet (2004); 11, Jobling (1994); 12, estimated from in vitro experiments reported by Giménez-Xavier et al. (2006); 13, estimated from experiments reported by Refstie et al. (2004); 14, estimated from experiments reported by Mente et al. (2003); 15, taken from experiments reported by Shearer et al. (1994); 16, taken from experiments reported by Bendiksen et al. (2006); 17, estimated from Moyes and West (1995); 18, taken from Halver and Hardy (2002).

pyruvate maintains low oxaloacetate levels and proper TCA cycle operation.

Control of rates in the TCA cycle

The rate of the TCA cycle is controlled by regulating the rates of acetyl-CoA and malate reactions. This is performed by a set of controllers that receive feedback information from ATP levels and regulate both the concentrations of acetyl-CoA and its flow rate into the TCA cycle simultaneously. Generally, in this model, each controller is operated in a process that may be compared with a valve: when it is fully open, conversion from one intermediate to another is promoted and vice versa. Each valve, denoted controller u_i, has its unique operational value, defined by a parameter α_i. At this value, rate of reaction is allowed at exactly 50% of its potential (the vertical lines in Fig. 3). The controllers (or valves) receive information about the value of the variable they regulate in what we call feedback information from a variable. Two types of controllers exist in this model: (*i*) a negative feedback controller, in which values $> \alpha_i$ cause the rate to decrease from 50% towards 0% (Fig. 3*a*), and (*ii*) a positive feedback controller that acts in the opposite direction (Fig. 3*b*). The magnitude of the deviation from the normal operational value, upon a feedback stimuli, is expressed as the parameter β_i. The higher β_i is, the more responsive the controller becomes to alterations in stimuli (Fig. 3).

For instance, suppose that rate of the reaction where acetyl-CoA joins oxaloacetate to form citrate is regulated by the negative feedback controller u_{citr} with information from ATP level. If ATP levels are higher than the normal operational value α_{citr}, then flow of acetyl-CoA into the TCA cycle will be lower than 50% of its potential. In practical terms, inhibiting the fueling component of the TCA cycle (for instance by inhibiting oxidation of fatty acids by closing the appropriate valve) will reduce the rate of the TCA cycle and consequently reduce the rate of ATP production. In this manner, the model maintains a consistent level of ATP.

Concentration of oxaloacetate inside the mitochondria should be kept at a low level even when amino acids are catabolized. This condition is satisfied because the reaction oxaloacetate to malate is favorable (recall that $k_{mal} \gg k_{TCA}$),

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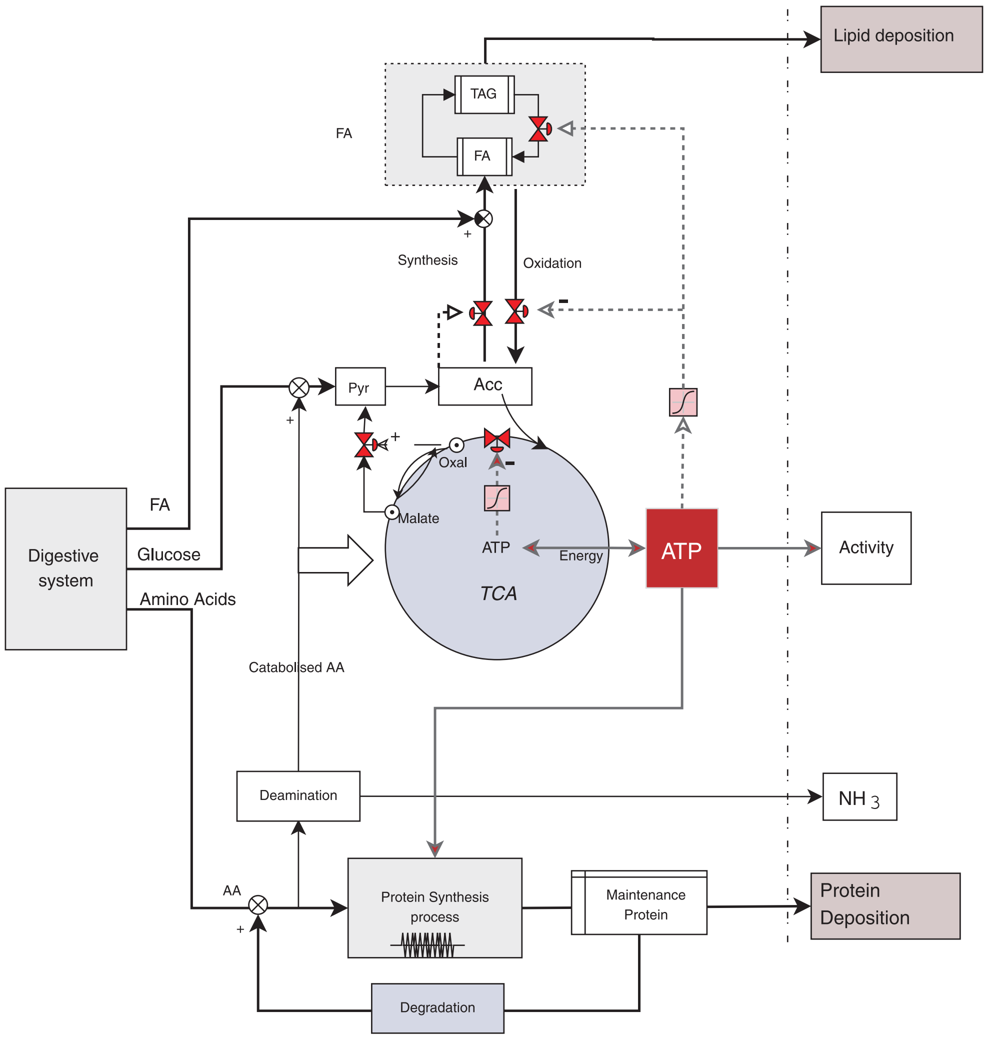
Fig. 1. Overview of the model. The inputs to the model are the digested nutrients fatty acids (FA), glucose, and amino acids (AA). The outputs are lipid deposition, NH_3, and protein deposition. Arrows correspond to material (mass) flow from one intermediate to another; metabolic processes are represented by shaded boxes; and broken lines represent rate control information. In this model, levels of ATP are used to regulate the flow rates to and from lipid synthesis and the TCA cycle.



with malate transported out of the mitochondria to the cytosol and consequently to pyruvate through the malate–pyruvate pathway. The flow rate of malate to pyruvate is controlled by the action of a positive feedback controller u_{oxal} such that higher oxaloacetate levels than the normal operation value (denoted α_{oxal}) promote flow of malate to pyruvate (Fig. 3*b*). The value of the parameters α_i and β_i are given in Table 2.

Protein synthesis and degradation

These are two energy consuming processes responsible for building protein tissues using available amino acids and degrading them. Protein synthesis is a complex process that incorporates numerous regulatory steps (Merrick and Hershey 1996; Kimball and Jefferson 2000; Merrick and Nyborg 2000). Only the translation process is modeled in the present study according to Bar and Morris (2007). The model consists mainly of initiation control and a continuous elongation model. The rate of protein synthesis is predetermined primarily by eukaryotic initiation factor (eIF)-2 and eIF-4 control mechanisms. eIF-2 is one of the central regulatory factors, determining the loading rate of the ribosomes on the mRNA according to the availability of amino acids. This process is a rapid one, occurring in a time scale of minutes (Murray et al. 1993), consuming energy in the form of GTP and ATP. The control of eIF-2 is complex and involves several factors that have the task of recycling eIF-2 after each ribosome loading. When availability of an amino acid decreases, eIF-2 is phosphorylated and consequently inhibits another prerequisite factor, eIF-2B, preventing it from recycling eIF-2 and inhibiting further loading of ribosomes (Hershey and Merrick 2000). The eIF-4 control mechanism is not included in the model development because its function primarily targets stress-related conditions other than AA deficiency (i.e., viral attack), which are well outside the scope and objectives of the model.

The newly synthesized protein is partially accumulated in the cells (often denoted in the literature as protein accretion), with the remaining utilized in tissues with rapid protein turnover, such as gut cells, and then degraded. Protein accretion represents the balance between protein synthesis and degradation (Carter and Houlihan 2001), but although

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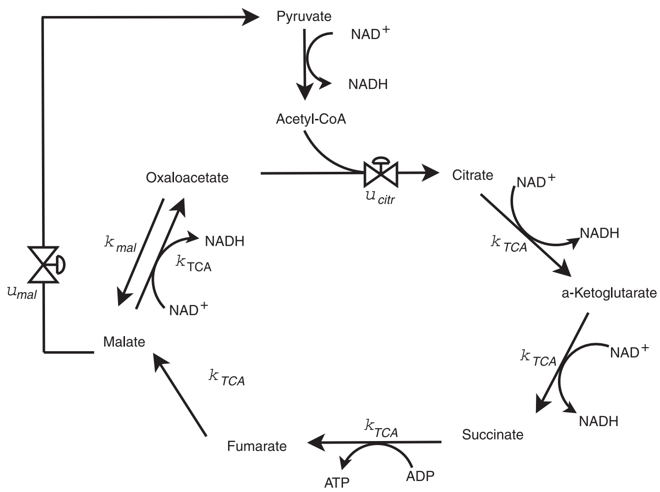
Fig. 2. Metabolic pathways modeled in the tricarboxylic acid (TCA) cycle in the present model. The two valves in the figure represent controlled reaction rate u_{citr} and u_{mal}; and k_{mal} and k_{TCA} are rate constants.



Houlihan et al. (1995) suggest measuring the rates of both protein synthesis and accretion, it is possible to calculate these using the modeled rate of protein degradation by measuring nitrogen excretion, thus enabling a more accurate calculation of the rate of protein accretion. Being dynamic, the model computes at each time step the concentration of nitrogen generated from amino acid breakdown and iteratively calculates the rate of protein degradation $\dot{p}_d$. Because this model is simplified and uses only one type of protein, we avoid the complexity arising from different rates of protein degradation in different tissues. The total protein in the body p_b is equal to the synthesized protein minus the degraded protein, or $p_b = p_{synt} - p_d$, and the changes of the variables with time can be written as

$$(4) \qquad \dot{p}_b = \dot{p}_{synt} - \dot{p}_d$$

Although $\dot{p}_{synt}$ is calculated using the model of Bar and Morris (2007), $\dot{p}_d$ is proportional to the protein mass in the body, or

$$(5) \qquad \dot{p}_d = v_d p_b$$

where the rate variable of protein degradation v_d is dependent on the age, size of the fish, and the temperature (Houlihan et al. 1995) and is modeled here as

$$(6) \qquad v_d = \frac{1}{\gamma}\frac{T}{T_{op}}\psi$$

where γ is a variable accounting for age and life-history stage, $T > 1$ is the temperature (in °C), $T_{op} \geq T$ is the optimal temperature for growth of a specific type of fish, and ψ is a parameter that describes the efficiency of the degradation rate for a given fish species. Note that γ is constant when simulating a short time span, hours and days, when age is approximately constant. Carter and Houlihan (2001) report that protein degradation is high at early life stages, thus γ is increasing with age. If seasonal changes are taken into account, then temperature and consequently v_d are time-dependent.

Amino acids resulting from protein degradation are partially re-utilized for protein synthesis, with the rest being catabolized. We denote the amino acids that are re-utilized to the protein synthesis process and those that are catabolized as f_p and f_t, respectively. Then, the degradation process generates amino acids that are diverted towards breakdown with fraction $f_t = 1 - f_p$. It is possible to estimate the value f_t (and consequently f_p) if measurements of nitrogen excretion from experiments with similar conditions are available. If measurements are unavailable, then f_t can be estimated using the protein turnover rate value, denoted in the literature as k_s (Houlihan et al. 1995). Alternatively, f_p can be estimated from data on amino acid retention if available.

Lipid biosynthesis and oxidation

Lipid metabolism incorporates the main energy storage and supply in the body. It is modeled as mass flow and transformation in a controlled manner (Fig. 4). Energy level regulates the process in several pathways through the action of the fatty acid oxidation and the rate of TCA cycle controllers, denoted as u_{xd} and u_{citr}, respectively. u_{xd} promotes supply of acetyl-CoA to the TCA cycle if energy is deficient, whereas u_{citr} promotes the entry of acetyl-CoA to the TCA cycle. Additionally, a negative feedback controller u_{ld} (lipid degradation) with information from ATP regulates the degradation of triacylglycerol (TAG) when ATP levels are low.

During fatty acid biosynthesis, acetyl-CoA is first transformed to malonyl-CoA and further to palmitate. The biosynthesis process is modeled through the following set of reactions:

$$(7) \qquad 7\ \text{acetyl-CoA} + 7\ CO_2 + 7\ \text{ATP} \xrightarrow{u_{ls}} 7\ \text{malonyl-CoA} + 7\ \text{ADP}$$

$$(8) \qquad \text{acetyl-CoA} + 7\ \text{malonyl-CoA} + 14\ \text{NADH} \xrightarrow{k_{ls}} \text{palmitate} + 14\ \text{NAD}^+$$

where k_{ls} is a rate constant of palmitate formation and u_{ls} regulates the rate of lipid synthesis (reaction 7) through a positive feedback with information from acetyl-CoA levels. Rate of TAG degradation is determined by the amount of available energy in the cell. The overall reaction can be described as

$$(9) \qquad \text{palmitate} \xrightarrow{k_{FA}} \text{FA} \xrightarrow{k_{TAG}} \text{TAG} \xrightarrow{u_{ld}} \text{FA}$$

where k_{FA} and k_{TAG} are rate constants of FA and TAG synthesis, respectively. The rates of TAG hydrolysis and β-oxidation are regulated by u_{ld} and u_{xd}, respectively, through negative feedback with information from ATP levels such that high ATP values inhibit transformation of TAG to FA and FA to acetyl-CoA and further accumulation of energy. Finally, the oxidation process can be described by the following reaction:

$$(10) \qquad \text{FA} + 7\ \text{NAD}^+ + 7\ \text{FAD}^+ \xrightarrow{u_{xd}} 8\ \text{acetyl-CoA} + 7\ \text{NADH} + 7\ \text{FADH}$$

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Fig. 3. Description of the controlled rates u_i: (*a*) a negative feedback control action, u_{citr}, u_{tag}, and u_{xd} for various β values; and (*b*) control of oxaloacetate levels by the action of positive feedback controller u_{oxal}.

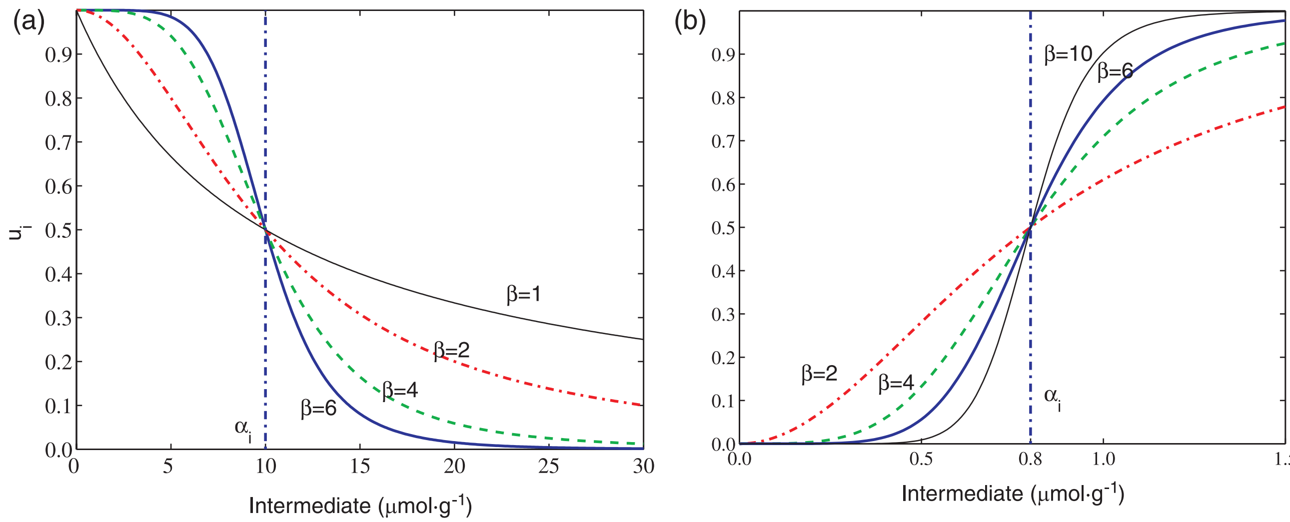
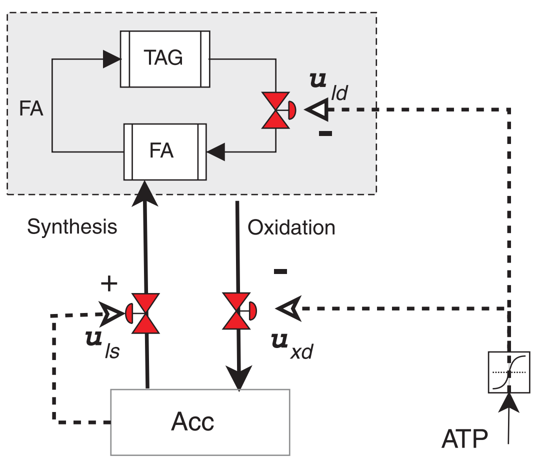


Fig. 4. Illustration of lipid metabolism model. The values u_{xd} and u_{ld} (illustrated as valves) determine the rates of fatty acids (FA) oxidation and triacylglycerol (TAG) breakdown according the ATP levels in the cells. Rate of lipid synthesis is regulated by u_{ls} according to acetyl-CoA (Acc) levels.



Energy expenditure, budget, and control

Among the most energy-consuming processes included in this model are protein synthesis, protein degradation, digestion, maintenance of ion gradients in the cells, and physical activity. The peptide elongation process is the principle source of energy consumption during protein synthesis. Here energy in the form of GTP can be systematically calculated as described in Bar and Morris (2007). The amounts of ATP and GTP required in the initiation process are negligible compared with those required in elongation. Protein degradation involving the protein ubiquitin is an ATP-dependent pathway, and through the ubiquitin–proteasome system, it requires about 2 ATPs for the breakdown of each peptide bond in the targeted protein (Varshavsky 2005). This system is believed to degrade the major contractile skeletal muscle proteins and plays a major role in muscle wasting (Attaix et al. 2005). This process is thus energy-demanding and central in energy expenditure in the fish. Because the ubiquitination is not yet fully understood, it is only partially modeled here, and although its kinetics are unknown, it is assumed that the time scale is in terms of minutes. Furthermore, ATP consumption can be easily calculated as the length of the peptide chain is known. Because the model presented in this paper focuses only on the metabolic pathways, energy consumption throughout the digestive system is simplified and implemented as a function of feed and temperature. Voluntary activity (associated with swimming) also plays a major role in energy consumption (Ballantyne 2001; Wood 2001) and can increase several-fold during exercise (Jobling 1994) to reach 90–120 μmol·(g dry mass)$^{-1}$·min^{-1} of ATP (Moyes and West 1995). Energy utilized for burst activity is considered insignificant in the context of aquaculture of Atlantic salmon. Fish reared in cages mainly swim at cruise speed, and thus only steady-state swimming is considered. The energy associated with the costs of blood circulation, endocrine function, osmoregulation, and other processes is termed essential.

There are essentially three sources of energy, i.e., lipids, amino acids, and glucose. Red muscle in salmonids contains up to 40% lipid, and β-oxidation is the dominant catabolic pathway to yield ATP for contraction (Kiessling et al. 2005). As mentioned earlier, to prevent large fluctuations of ATP, ATP levels in cells are tightly regulated using several flow controllers. When the level of ATP is exceedingly high, the controllers calculate the correct flow that leads to a decrease in ATP levels, and vice versa. Activity level and, consequently, ATP consumption has a major effect on the flow of FA into and out of the TAG reserves. Amino acid catabolism is another process that plays an important role in ATP formation. The model calculates the amount of ATP derived at each instance from amino acid catabolism processes, whereas the energy controllers adjust the rates of β-oxidation and TCA cycle to maintain ATP levels within an acceptable range.

Growth of whole body

In the present model, we assume that body mass is mainly composed of proteins in muscle tissues such as myofibrillar

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and actin, TAG in the adipose cells, water, and ash. This assumption generally fits the model for fish (Shearer et al. 1994). The total mass can be written as

$$(11) \quad m_w(t) = w(t) + p(t) + l(t) + \zeta(p(t) + l(t))$$

where $m_w(t)$ is total wet body mass, $p(t)$ is the mass of protein deposits, $w(t)$ is the mass of water, and $l(t)$ is TAG, all in units of grams (Olsen and Balchen 1992). ζ is the fraction of ash in the dry matter, assumed constant. Data sets (Aksnes et al. 1986; Shearer et al. 1994) suggest that 3.8 g water are added to the total body mass for each gram of protein deposited (this value is denoted as ρ_p). Additionally, ρ_L = 0.7 g water are reduced for every additional gram of lipid in the adipose cells. Mass of water can be described as

$$(12) \quad w(t) = \rho_p\, p(t) - \rho_L\, l(t)$$

Starvation

During starvation, amino acids are not available from food intake, and protein deposited in the body must be degraded to supply the required amino acids for synthesis of essential proteins. Reduced fractional rates of protein synthesis during starvation have been found in almost all animals, including fish, which implies that turnover rates are lower in starved fish (Houlihan et al. 1995). In this model, only protein degradation is slowed down actively, whereas reduced rates of protein synthesis are an outcome of initiation control due to a limited supply of amino acids, which yields a reduction in the total turnover rate. Lipid reserves are used to supply energy. When those are depleted, the rate of protein degradation is increased to meet the energy demands.

Choice of parameters

Most of the parameters used in the model were estimated using data from previous published studies (Table 2). In the literature, data on steady-state values of ATP vary: Waiwood et al. (1992) and Caldwell and Hinshaw (1994) report 6.93 and 6.5 μmol·g^{-1} in the white muscle of Atlantic salmon and rainbow trout, respectively, Thillart et al. (1976) report 10.22 μmol·g^{-1} in dorsal white muscle of goldfish, and Erikson et al. (1997) report ATP values as high as 11.8 μmol·g^{-1}. Studies of kinetic rates inside the TCA cycle date back to the 1930s and 1950s and are well documented in the literature (Table 2). A substantial number of parameters involving protein synthesis and degradation are unknown and need to be estimated or assumed, including γ and ψ. Parameters involved in eIF-2 initiation control are largely taken from yeast cell values. This, however, should have little or no impact on growth because of the short operation time of initiation control (seconds to minutes) relative to growth (weeks to months).

Results

Refstie et al. (2004) have conducted experiments on Atlantic salmon using four different diets. In these experiments, low temperature (LT) fish meal (FM) was replaced by fish protein hydrolysates (FPH) in increments, producing diets containing 0% (FM), 5% (FPH-05), 10% (FPH-10), or 15% (FPH-15) FPH so that the amino acid composition of the diet differed. In this paper, we present only results from the FM and FPH-15 diets (Table 3).

Table 3. Free amino acids composition in diets, taken from Refstie et al. (2004).

	Diet	
Amino acid	FM	FPH-15
Arg	6.72	6.78
His	2.62	2.31
Ile	4.15	4.19
Leu	7.20	7.22
Lys	7.16	7.11
Met	2.58	2.63
Met + Cys	3.67	3.72
Phe	4.04	4.14
Phe + Tyr	7.25	7.40
Thr	4.38	4.41
Val	5.10	5.05
Ala	5.92	6.08
Asp + Asn	9.05	9.25
Cys	1.09	1.09
Glu + Gln	14.75	15.43
Gly	6.51	7.21
Pro	4.93	5.35
Ser	5.03	5.31
Tyr	3.21	3.26

In the experiment, 16 groups of 60 fish per group (m_w = 163 g) were fed for a period of 68 days. The water temperature (T) during the experiment period was 8.3 ± 0.5 °C, and O_2 saturation of the outlet water ranged from 81% to 97%. The feed intake was measured as 1.163% body weight (denoted m_w) per day. The feeding regime was further divided into four periods, and the feed intake was measured during these periods.

To estimate the parameters of protein degradation and maintenance metabolism, the model was simulated with the FM diet. Refstie et al. (2004) report nitrogen excretion owing to nitrogen metabolism of approximately 13 g·kg^{-1} gained, which resulted in $\hat{NH}_3$ = 2.08 g = 0.122 mol ammonia excreted during the 68 days. The parameters γ and ψ in eq. 6 are corrected to fit the final fish mass of 323 g. After estimating the parameters, other diet compositions (e.g., FPH-15) and environmental conditions could be tested without altering any of the parameters in the model. The above data set was simulated using MATLAB (The Mathworks Inc., Natick, Massachusetts) and C programming language.

Comparison

The results of the simulation are compared with the results reported by Refstie et al. (2004) to examine the error, i.e., predicted values less the reported values. Predicted protein content in the body using the FM diet was calculated to be 15.5% of body weight (Table 4), whereas Refstie et al. (2004) report 12.7% (sum of dehydrated amino acids, excluding tryptophan). Predicted energy content in the body was 8.56 MJ·kg^{-1} versus a reported value of 8.2 MJ·kg^{-1} (calculated using 23.6 kJ·g^{-1} protein and 39.5 kJ·g^{-1} lipid). Estimated growth rate during the first period of the experiments was higher than that reported by Refstie et al. (2004)

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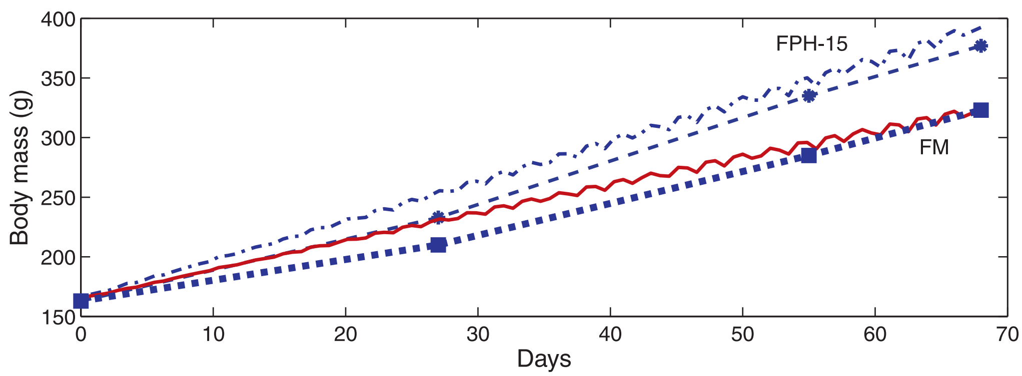
Table 4. Comparison of the predicted results from the model with results reported by Refstie et al. (2004).

	FM		FPH-15	
Diet	Predicted	Experiment	Predicted	Experiment
Growth (g m_w)	323.90	323	392.40	377
Specific growth rate (SGR)				
0–27	1.25	1.02	1.60	1.43
28–55	0.87	1.22	1.12	1.45
58–68	0.64	0.97	0.85	0.90
0–68	1.00	1.09	1.29	1.33
Thermal growth coefficient (TGC) (0–68 days) × 1000	2.48	2.67	3.29	3.36
Protein (% m_w)	15.50	12.70	15.80	13.10
Lipid (% m_w)[a]	11.50	10.38	12.10	12.53
Water (% m_w)[b]	70.90	70.50	70.10	68.10
Energy ($MJ \cdot kg^{-1}$)	8.56	8.20	8.90	9.10

Note: The values SGR and TGC are computed as described by Refstie et al. (2004). m_w, total wet mass of the body.
[a]Lipid content in the results from experiments is calculated from crude protein by S. Refstie (personal communication, 2007). Lipid content was found to be 13.18% for FM diet and 15.22% for FPH-15 when calculated from protein as sum of amino acids.
[b]Water in experiments is from S. Refstie (personal communication, 2007)

Fig. 5. Prediction of growth of salmon using the dynamic model compared with the results from experiments reported by Refstie et al. (2004) for two diets: FM and FPH-15. Predicted FPH-15, dotted–dashed line; reference FPH-15, dashed line, solid circles; predicted FM, solid line; and reference FM, dotted line, solid squares.



(Fig. 5), with maximum deviation between the modeled and experimental results at around day 30. We note that the error of the final specific growth rate (SGR) value is small. Results from using the FPH-15 diet also demonstrate a large error in growth estimates during the first period. However, the error is diminished with time. The final predicted mass was 392.4 g, with an error of 4.1%. Predicted body composition was calculated to be 15.8% protein (sum of amino acids), 12.1% lipid, and 70.1% water, with a predicted energy content of 8.9 $MJ \cdot kg^{-1}$.

Variations of body composition are presented (Fig. 6*a*) and compared with the reported results (Fig. 7). Mean O_2 consumption during this period was about 9 $\mu mol \cdot (g\ dry\ mass)^{-1} \cdot min^{-1}$. However O_2 consumption was not measured in the experiment and this value could not be compared.

Predicted amino acid retention was generally lower than that reported by Refstie et al. (2004), particularly cysteine retention (Table 5). Cysteine has low digestibility (reported about 67% digestibility) and part of the cysteine retained in the body is synthesized from methionine and serine. Both nitrogen excretion and oxygen consumption increased linearly during the simulation time (Fig. 6*b*).

Case study: testing limiting amino acids

The model can serve as a tool to investigate which amino acids in the diet are growth-limiting. In their experiments, Refstie et al. (2004) used marginal concentrations of lysine, methionine, and cysteine in the FM diet. Assuming that the measured digestibility was correct, lysine was the limiting amino acid, whereas methionine had marginal concentrations in both the FM and FPH-15 diets, because part of the methionine, in conjunction with serine, is utilized for synthesis of cysteine through the transsulfuration pathway, reducing availability of the methionine for protein synthesis. Thus, methionine is a potential limiting AA, and increasing lysine will not yield a significant increase in growth. To test this hypothesis, the model was run using varying lysine and methionine concentrations in the diet while recording final growth and lipid composition (Fig. 8). When lysine in the diet was increased above its initial value of 7.16% while methionine was held constant at 2.58%, no increase of m_w above 323 g was registered. This is apparent in Fig. 8*a* where body mass remains in the same contour area, 323 g, for these values. Body fat deposition, however, was higher in the above case, as can be expected, because the remaining

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Fig. 6. Results of simulation of body composition using standard diet FPH-15: (*a*) body composition as percent of total body weight (notice the double axes, left for protein–lipid and right for water); and (*b*) NH_3 excretion (solid line) and O_2 consumption (dotted line) (in $g \cdot m_w^{-1}$).

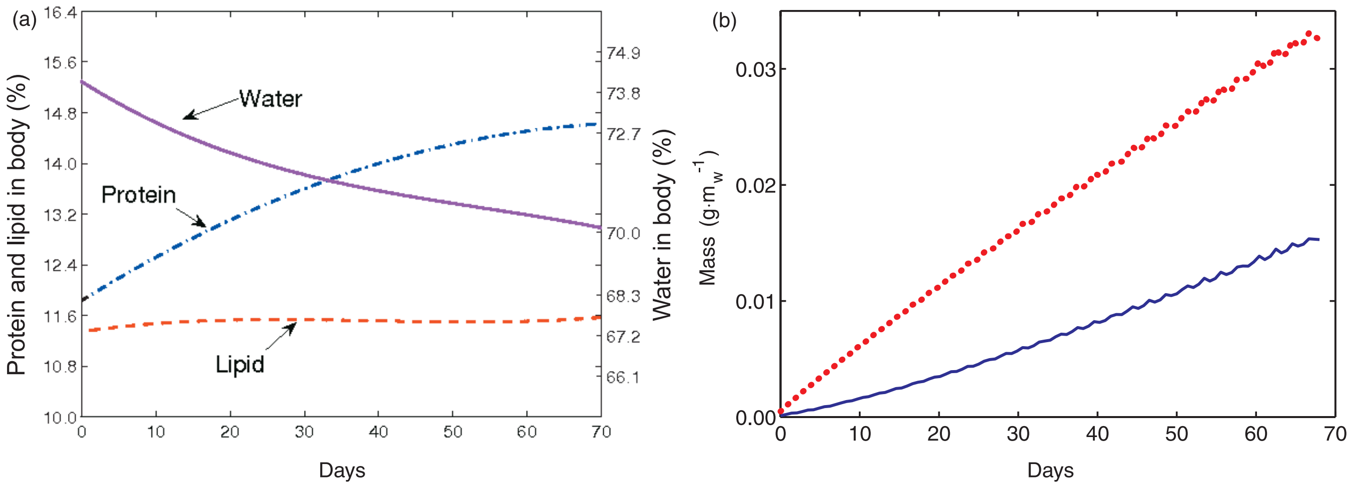

Fig. 7. Comparison between predicted body composition (solid bars) and the experiments (open bars) reported by Refstie et al. (2004) for two diet compositions: FM and FPH-15. Water composition in the experiments is estimated.

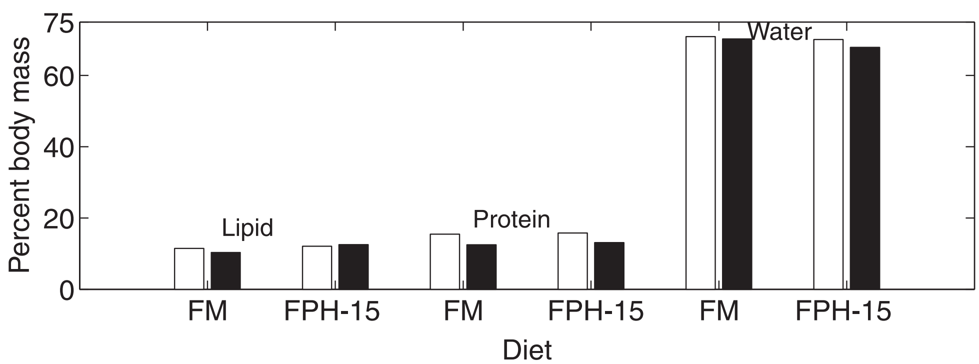

Table 5. Comparison of some predicted amino acid (AA) retention (% AA digested) with AA retention from experiments reported by Refstie et al. (2004).

	FM		FPH-15	
Diet	Predicted	Experiment	Predicted	Experiment
Arginine	48.05	44.6	48.1	49.4
Lysine	61	65.9	62	71.5
Threonine	54	61.6	54.2	65.6
Methionine	60	66.2	59.5	71.6
Histidine	48.9	55.3	55.2	67.3
Cysteine	60.5	82.1	61.3	81.7

amino acids that were not utilized in protein synthesis were diverted toward catabolism. This inhibited oxidation of fatty acids as long as the energy gain from amino acid catabolism was high. An equivalent example is given for the case of a constant dietary lysine concentration. When lysine was held constant at 7.16%, methionine values above 2.59% did not enhance growth above 324 g. This phenomenon could be predicted by Fig. 8*a*, as this methionine value did not cross any horizontal line.

However, in the case in which both methionine and lysine were increased (2.65% (additional 0.07%) and 7.32% of the dietary protein, respectively), the model predicted an increase of 7 g in growth (Fig. 9) and a decrease of 0.4% in fat deposition (Fig. 8*b*).

Following the analysis above, we conclude that an optimal weight gain through increases in lysine and methionine can be obtained by balancing amino acids (according to the diagonal broken line in Fig. 8*a*).

Discussion

This model was intended to help in understanding the mechanisms that determine the distribution of body composition in fish, both qualitatively and quantitatively. The model can increase our understanding of how intracellular energy levels, growth, and body composition are related. Additionally, because the model can operate in a wide range of conditions, it can be used as a tool to discover bottlenecks in fish feed. For instance, we demonstrated how various amino acids are coupled so that increasing one amino acid without the other does not lead to a significant improvement in growth.

The model

Because the model is relatively complex, not all the parameters are available. A few unknown parameters can be estimated by measuring nitrogen excretion, whereas others can be estimated by simulating an existing experiments with

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Fig. 8. Variations in (*a*) body wet mass (m_w) and (*b*) percent lipid in the body as a function of amino acids lysine and methionine. The broken line in (*a*) suggests optimal combination of the two amino acids, which results in maximum body weight.

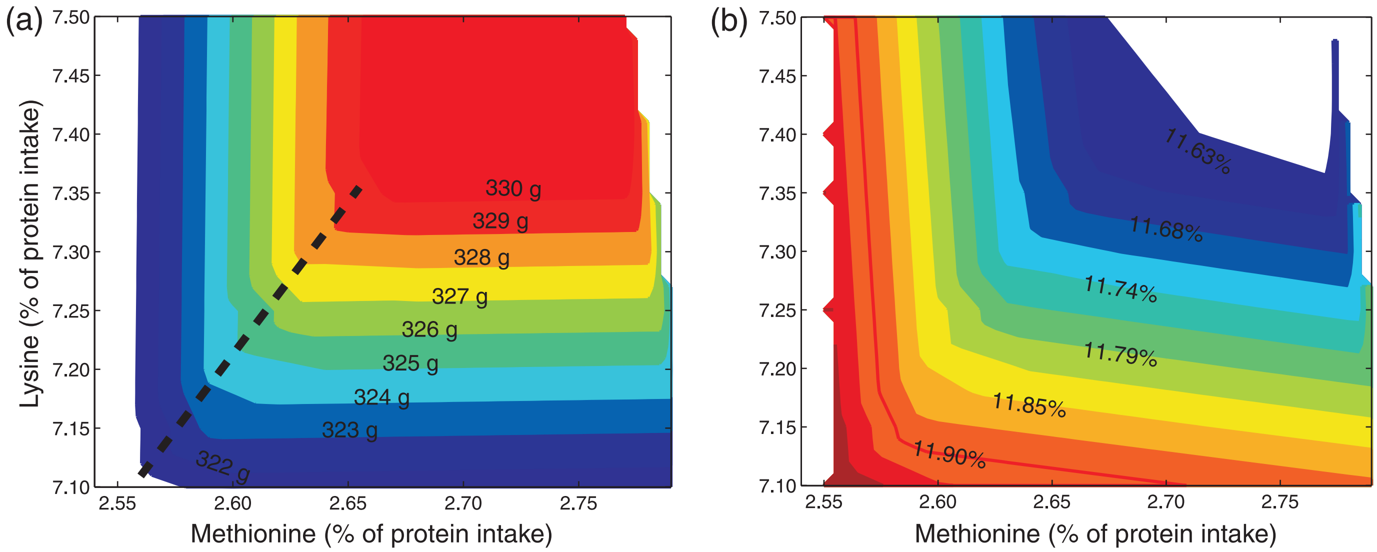
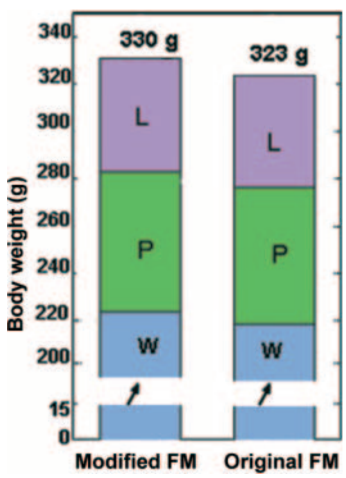


Fig. 9. Comparison of water (W), protein (P), lipid (L), and total body weight between the original diet FM (right) and the diet modified with additional 0.07% methionine and 0.16% lysine (left). The *y* axis was rescaled for brevity of presentation.



similar biotic and environmental conditions. Because the model is only a simplified description of the real metabolic system, prediction errors (deviations of predicted results from experiments) are unavoidable. The errors can be diminished under certain conditions by using measurements to update the dynamic model, but unfortunately, not many variables can be updated in real time using measurement technology available today. In fact, the only variables that can be obtained within a reasonable experimental cost are the output variables such as body mass (m_w), O_2 consumption, and CO_2 and NH_3 excretion. It is also possible to measure body composition upon slaughtering the fish, but this method requires expensive large-scale experiments, as updates should be acquired at a high sampling rate. By applying nonlinear observer rules (see, for example, Gelb 1994), it is feasible to update the variables such that prediction error diminishes. Note that additional variables can be measured using nondestructive methods such as cannulation and nuclear magnetic resonance spectroscopy (Thillart and Raaij 1995) to improve the estimates, but these methods are usually expensive and complicated. In the absence of measurements, feedback information is unavailable and prediction errors may accumulate with time, rendering the model unreliable for long-term predictions, e.g., years.

Amino acids synthesis is omitted (except cysteine and alanine) based on the assumption that all nonessential amino acids are supplied in sufficient amounts with the diet. This assumption can introduce inaccurate estimation of nitrogen excretion in cases of high rates of amino acid synthesis. Although the fish re-utilize nitrogen in amino acid synthesis, the model does not incorporate this process and may overestimate nitrogen excretion. This might explain the lower values for predicted vs. measured retention, i.e., the fraction of amino acids diverted to breakdown was overestimated. Decreasing f_t, however, may cause overestimated gain of body mass as higher protein retention implies larger protein and water fraction deposited.

An important characteristic of the model presented here is modularity. The metabolic model can be extended by designing other submodules and interconnecting them to the existing model without the call for major modifications. Its structure enables easy input–output links of flow between submodules. Including additional pathways (e.g., amino acids synthesis, urea cycle, and hormonal balance and regulation in the body) inside the metabolic model is possible, yet because a large portion of the dynamics and rates are still unknown, this may introduce unnecessary complexity, which may increase the prediction errors. Simplifying the model eliminates these unknowns, but unmodeled dynamics might risk divergence of the variables from the real values.

The experiments

Generally, the predicted values in the model were in agreement with results from experiments. The experiments reported by Refstie et al. (2004) demonstrated low SGR in the first 10 days in contrast to other experiments conducted under similar conditions (Austreng et al. 1987). This is in

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accordance with simulations that showed a higher predicted SGR for the same period of time. In their discussion, they mention the lack of acclimation as the cause for the low SGR, as exposure to stress resulted in low feed intake and growth repression. Most likely, the 3%–4% error in prediction of growth might reflect the lack of acclimation and could be diminished otherwise.

Predicted protein in body composition was found to be 2.7% higher than the reported, although the reported value did not include the amino acid tryptophan, which would otherwise reduce this deviation. A logical explanation for the high amino acid content predicted might be an inaccurate rate of protein degradation. However, the relatively low amino acid retention values predicted do not support this hypothesis as a higher degradation rate would further decrease the retention values. Inaccuracy in amino acid requirement in the model might be another reason for the deviation between predicted and reported value.

S. Refstie (AKVAFORSK Sunndalsøra, Norway, personal communication, 2007) calculated lipid content from crude protein (CP) and total energy content in the body to be 12.53%. Prediction of lipid content in the model was in agreement with the reported results from CP with no more than 1% difference. However, 3% higher lipid content (15.22%) was predicted by the model when protein was regarded as the sum of amino acids.

Application

There are several applications of the model. By simulating various feed compositions, amounts, and regimes, the model can be utilized to compare the efficiency of several intake strategies. The model facilitates studying the effects of changes in feed composition on the first limiting amino acids and body composition. For instance, we demonstrated how lysine and methionine were coupled when testing for limiting amino acid effects using the FM diet. In fact, cysteine is also coupled with methionine as the former can be synthesized from methionine using serine, if cysteine is supplied in low quantities in the diet. In simulating several intake variations of the FM diet, it was found that the reported value of 2.58% methionine was insufficient to both produce protein and synthesize cysteine, and thus methionine is a strong candidate as a limiting amino acid, particularly as cysteine has low digestibility (Sorensen et al. 2002). The FM diet presented by Refstie et al. (2004) contained approximately 1.09% cysteine with digestibility of 63%, which implies insufficient cysteine uptake and the latter should be synthesized from methionine. The content of methionine in the FM diet, however, was marginal and even its smallest transformation to cysteine rendered it a limiting amino acid. To improve the balance of amino acids in the diet, one can use the model to investigate incremental changes in amino acid composition. It was shown that to increase growth and retention of amino acids while decreasing lipid deposition, both lysine and methionine should be increased simultaneously. Thus, according to the model, growth was expected to improve if lysine would be increased to 7.32% in the FPH-15 diet. Figures of this type can be generated for any pair of amino acids and used as tools to predict and design the limiting amino acid and its effect on mass and body composition.

Quantifying the nutrition pathways helps solve the question of which source of ATP the fish uses to satisfy energy demands for activity. Because the tissues have a limited capacity to store amino acids, any excess is oxidized, releasing energy regardless of its demand. If energy demand is low, the controllers decrease the rate of lipid oxidation to maintain energy levels within acceptable margins. In a balanced diet, rate of breakdown of amino acids is low; therefore glucose and lipids are utilized as an ATP source by the energy controllers to satisfy energy requirements.

Future work

The metabolic pathways presented in this model give an incomplete description of the nutrient metabolism in the body. Additional submodels can be incorporated into this model at both the microlevel, such as the pentose–phosphate pathway and amino acid synthesis, and the macrolevel, such as digestibility, appetite, and behavior. Better measurement systems will increase the accuracy of prediction, particularly for simulations with long periods of time. As the model improves, we would like to co-operate with researchers who are interested in fitting the model to other species, environments, or experiments. This will further increase its accuracy, which might make it a potential tool in future research within aquatic sciences. We note that the complete set of model equations are available in Bar (2007).

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